Predicting Stock Movements Using Twitter Sentiment Analysis

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Agenda

1. Motivation/research question
2. Modeling approach
3. Data acquisition
4. Analysis plan
5. Bias and uncertainty
6. Results
7. Next steps

Motivation/Research Question

- Motivation
  - We were trying to connect public sentiment toward a stock to its market behavior by analyzing tweets that contained stock ticker symbols
  - We were especially curious about whether or not social media has a real impact on movement since stock trends are largely dependent on the people
- Research Question
  - Can natural language processing of tweets containing stock ticker symbols (e.g., $AAPL) be used to forecast next-day market performance?

Modeling Approach

- **Hypothesis:** We hoped to build a 85%+ accurate model that could predict a stock's next day trend of the closing price compared to the day prior
- Modeling Approach
  - We hoped to build a 85%+ accurate model that could predict a stock's next day trend of the closing price compared to the day before by doing the following:
    - Collect market and Twitter data
    - Use the VADER package to calculate an average sentiment for tweets each day
    - Building a supervised learning model where the stock price next day is the target and features include the price difference and sentiment score
    - Start with linear relationships and then do tests to capture non-linearity

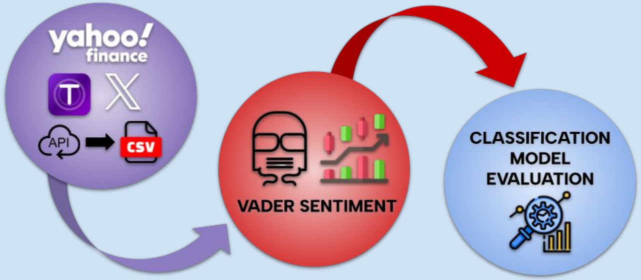
Data Acquisition

- Stock data from yfinance python package
  - Pulls from Yahoo Finance
- Twitter data from twitterapi.io
- Queried each day from October 1, 2020 to October 1, 2025
- Cleaned and combined both datasets to create our final dataset
  - See Data Dictionary (right)

Data Dictionary

date (datetime)	Date to analyze
average_sentiment (float)	Average vaderSentiment score of 25 tweets for the given date
close (float)	Closing price of given date
open (float)	Opening price of the next day
price_difference (float)	open minus close

Analysis



Plan

- Use **Vader Sentiment** to convert tweets into numerical data. Find the average sentiment for each day’s tweets, and aggregate that with the price difference between close and open. Use this data to train various models, and compare performance. Evaluation metrics include accuracy, precision, recall, F1 Score, and confusion matrix.

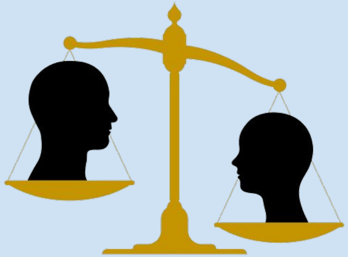
Justification

- VADER is really strong on capturing both formal and informal language, which is perfect for social media. Also, the use of multiple models ensures we’re accounting for both linear and non-linear relationships between our data. With this, we feel we are able to optimize our results.

Tricky Analysis Decision

- When would we look to analyze?
  - Active trading period? (9:30am - 4:00pm)
  - Inactive period?
  - Both?
- Focused on the downtime between days to focus on the difference between that day's closing price, and the next day's opening
  - Better reflected twitterapi.io's response structure, receiving tweets from later in the day
  - Limitations with the Twitter API

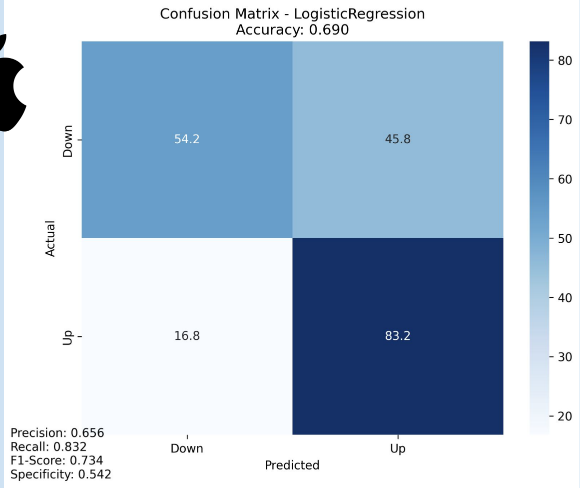
Bias and Uncertainty Validation

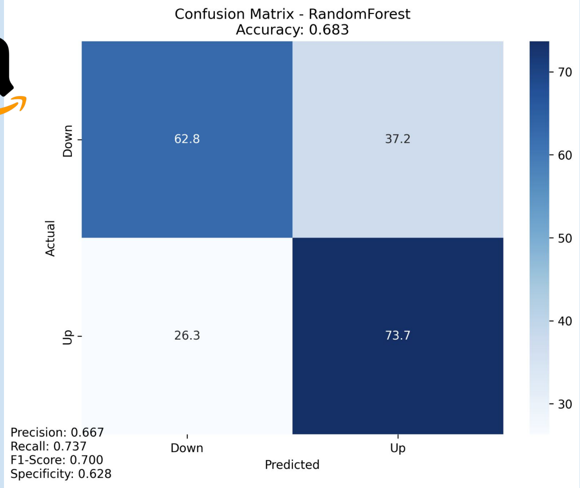


- Selection Bias:
  - Only tweets **containing explicit stock ticker symbols** (e.g., $AAPL) are collected. This excludes informal mentions of the stock, which might represent a large share of market sentiment.
  - Correction: We plan to expand keywords to include company names and hashtags (e.g., “Apple”, “#AAPL”) to reduce underrepresentation.
- Engagement Bias:
  - Tweets with **high likes or retweets may be overrepresented** in the dataset or disproportionately influence average sentiment, even if they are not representative of the broader community.
  - Correction: We normalize sentiment scores by engagement volume and limit the impact of outlier tweets.
- Time Bias:
  - Tweets are collected **after market close**, which may miss intra-day reactions or news-driven sentiment shifts.
  - Correction: We consider multiple collection windows (e.g., 12h, 24h before close) and analyze temporal effects.

Results and Conclusions

- AAPL
  - Best Model: Logistic Regression
- AMZN
  - Best Model: Random Forest
- Similar Classification Statistics
  - ~70% Accuracy
  - ~0.66 Prediction
  - ~0.7 F1-Score
  - ~0.55-0.6 Specificity
- Prediction was decent, but not perfect
  - Other models performed slightly worse
- Are movements purely influenced by the opinions of the people on social media? Likely not.





Next Steps

- Getting Twitter data was costly
  - Retrieving tweets for one stock would take hours
  - Difficult to generalize to other stock given our timeline and resources
- Accuracy was unideal, likely because movement is dependent on other social media discourses as well
  - There are other features that goes into people’s decision to buying/selling:
    - News
    - Personal constraints
    - Corporate actions
    - Insider activity
    - Etc.

References

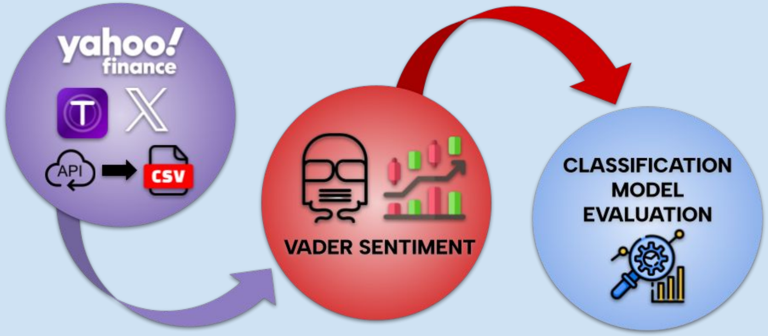
[1] TwitterAPI.io, “Twitter API — Real-time and historical social data,” TwitterAPI.io. [Online]. Available: https://twitterapi.io/. [Accessed: Oct. 17, 2025].

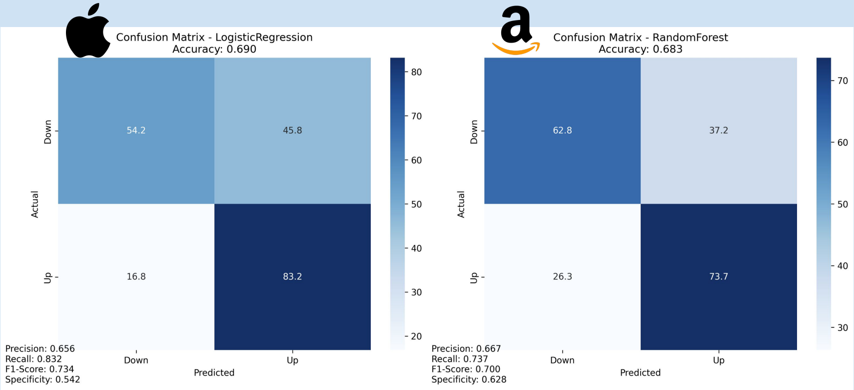
[2] Yfinance, “yfinance: Yahoo! Finance market data downloader,” PyPI, 2025. [Online]. Available: https://pypi.org/project/yfinance/. [Accessed: Oct. 17, 2025].

[3] C. Hutto and E. Gilbert, “VADER: Valence Aware Dictionary and Sentiment Reasoner,” PyPI, 2025. [Online]. Available: https://pypi.org/project/vaderSentiment/. [Accessed: Oct. 10, 2025].

[4] P. R. de Lima, T. R. da Silva, and L. Pereira, “Twitter sentiment and stock market movements: The predictive power of social media,” Centre for Economic Policy Research (CEPR), VoxEU Column, 2020. [Online]. Available: https://cepr.org/voxeu/columns/twitter-sentiment-and-stock-market-movements-predictive-power-social-media. [Accessed: Oct. 10, 2025].

Thank you!





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